Agent S2: A Compositional Generalist-Specialist Framework for Computer Use Agents

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Paper version: arXiv 2504.00906v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *they often underperform compared to specialist models in domain-specific subtasks*

One generalist model plans, acts and grounds pixels; grounding and replanning stay the bottlenecks.

Three limits of monolithic agents

1. **Grounding slips:** text targets land on wrong pixels.
2. **Long horizons:** pop-ups defeat a fixed plan.
3. **Reactive replanning** fires only after failure.

Evaluation setup

OSWorld · 369 tasks | WinAgentArena · 154 | AndroidWorld · 116 | screenshots only | 15 & 50 step budgets | ablation: 65-task set

- **All online**, functional checks; no offline step accuracy.
- **Held-out OSes:** Windows and Android; worker-only on AndroidWorld.

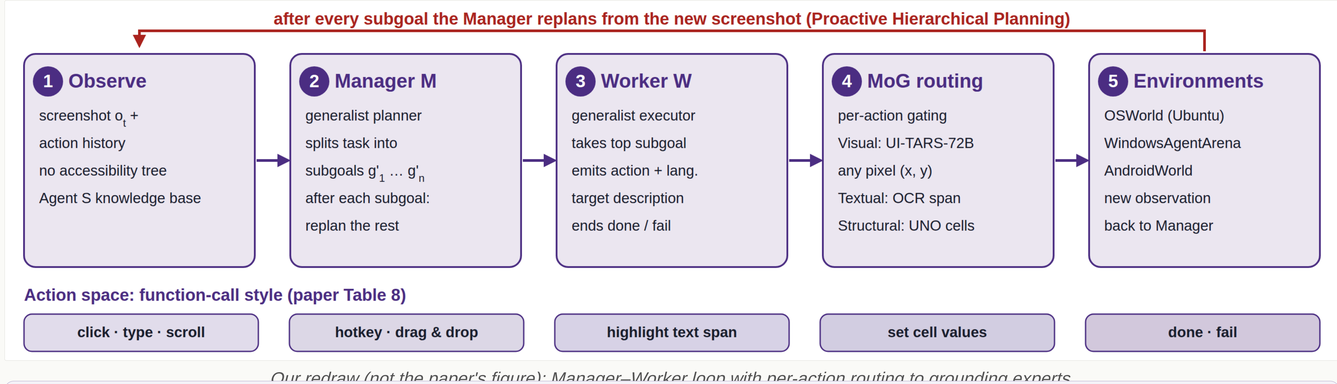
Our read (CUA Radar)

61	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; held-out OSes; strong baselines; matched 15/50-step budgets; ablation; code.
- **Flags:** variance and cost not reported; 65-task ablation subset.
- Pattern C10, planner-grounder split. Authors' numbers, not rerun.

Method: generalists plan and act, specialists ground

- **Sees / acts:** screenshot + action history in; function-call actions out (click, type, scroll, set_cell_values…).
- **MoG:** Worker routes each action + language target to Visual (UI-TARS-72B), Textual (OCR) or Structural (UNO) expert.
- **PHP:** Manager regenerates the remaining subgoals after **every** subgoal, from the latest screenshot.

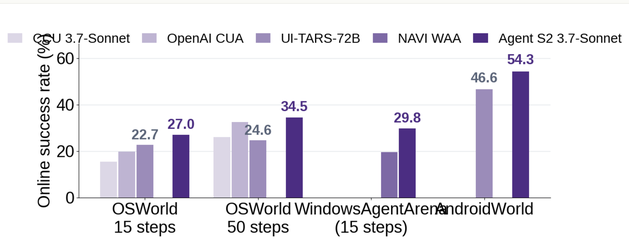


Our redraw (not the paper's figure): Manager–Worker loop with per-action routing to grounding experts.

$$a_t \sim \pi_W(g_i, o_t) \;\rightarrow\; G_j(d_t) = \langle x, y \rangle$$

Worker policy emits an action and target description d; the routed expert G_j returns coordinates (paper §3.1)

Key results: online success rate (%)



Tables 1/3/4, redrawn; all online. UI-TARS-72B = DPO on OSWorld, SFT on AndroidWorld.

27.0 / 34.5 OSWorld · 29.8 WAA · 54.3 AndroidWorld

SO WHAT → *New SOTA as reported on all three; same stack on every OS; NAVI also reads the a11y tree.*

Ablation: MoG on a 65-task OSWorld subset

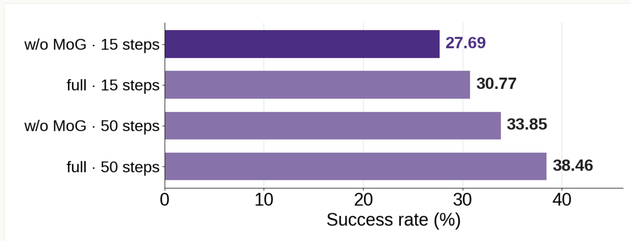


Figure 5 (text values), redrawn: success rate with and without Mixture-of-Grounding.

SO WHAT → *Proactive planning adds +4.62 / +6.15 pts over reactive (15/50 steps); dropping the textual expert lowers subtask SR 70.6 → 65.2, structural 73.7 → 69.4. Small open experts (UI-TARS-7B, UGround-7B) beat Claude-3.7-Sonnet in the visual slot.*

Headline Numbers

27.0%
OSWorld 15-step success
vs 22.7 UI-TARS-72B-DPO; 50-step: 34.5 with the same stack

- **34.5%** OSWorld 50-step
- **29.8%** WindowsAgentAre
- **54.3%** AndroidWorld

The paper's own controls

CONTROL	AGENT S2	COMPARED WITH
Steps 15 vs 50	27.0 vs 34.5	CUA 19.7 vs 32.6
Backbone 3.5 (new)	24.5 / 33.7	CCU 14.9 / 22.0
Input	screenshots only	Agent S: tree + shots
Drop expert (subtask)	text 70.6 → 65.2	struct 73.7 → 69.4

Takeaway

Language targets, specialist grounding, replan after every subgoal: reported SOTA on three online benchmarks, with prompted models and no training.

OSWorld 27.0 / 34.5 | WAA +52.8% rel | AndroidWorld 54.3 | prompted, no training

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Paper | Code